Semi-Supervised SAR ATR Based on Contrastive Learning and Complementary Label Learning

Chen Li, Lan Du, *Senior Member, IEEE*, and Yuang Du, *Graduate Student Member, IEEE*

***Abstract*— Deep-learning-based methods have recently achieved significant advancements in synthetic aperture radar automatic target recognition (SAR ATR). However, these methods typically rely heavily on extensive annotations, which are difficult to obtain for SAR images. Semi-supervised learning offers a solution to improve model performance with limited labeled data by leveraging unlabeled data. The mainstream semi-supervised learning methods for SAR ATR typically select high-confidence unlabeled images to assign pseudo-labels for their inclusion in the model training process. However, the large number of low-confidence unlabeled images are not efficiently utilized. To address this issue, a semi-supervised SAR target recognition method based on contrastive learning and complementary label (CoL) learning is proposed. First, CoL learning assigns CoLto low-confidence unlabeled images based on their minimum prediction probabilities. Subsequently, a threshold is set to filter out unreliable CoL, thereby mitigating the adverse effects of erroneous CoL. This approach ensures the effective and comprehensive utilization of low-confidence unlabeled images. Additionally, we propose a contrastive loss that incorporates CoL. Compared to traditional contrastive losses, our proposed contrastive loss constructs a richer set of negative sample pairs by leveraging the characteristics of CoL more effectively. Consequently, this approach improves the utilization of low-confidence images and further improves recognition performance. In contrast to the current state-of-the-art semi-supervised recognition methods, experiments on the MSTAR dataset demonstrate the better recognition performance of our proposed method with limited labeled images.**

***Index Terms*— Complementary label (CoL) learning, contrastive learning, synthetic aperture radar (SAR), target recognition.**

I. Introduction

IN RECENT years, synthetic aperture radar (SAR) automatic target recognition (ATR) has emerged as a pivotal tool for image interpretation across diverse domains, including military and civilian applications [1], garnering significant attention. Due to the strong feature representation abilities, deep learning methods have been widely utilized in SAR ATR, resulting in notable advancements. Nonetheless, deep learning methods are dependent on a data-driven paradigm, necessitating a substantial quantity of labeled images. Compared to optical images, the distinct imaging mechanism of SAR images poses challenges for visual interpretation, thereby limiting the availability of annotations for SAR images. Using a limited number of labeled images can lead to overfitting, whereas a considerable amount of unlabeled images remain unused, ultimately constraining recognition performance.

In addition to utilizing a limited quantity of labeled images, semi-supervised learning [2], [3] can improve recognition performance by leveraging a significant quantity of unlabeled images for model training. Current mainstream semi-supervised learning methods primarily rely on two techniques: pseudo-labeling [4] and consistency regularization [5], [6]. The objective of pseudo-labeling methods is to iteratively select unlabeled images with high prediction confidence for inclusion in network training, thus effectively utilizing unlabeled images. The core idea of consistent regularization lies in preserving the invariant properties of unlabeled images across diverse perturbations, thereby encouraging the model's generalization capabilities.

In recent years, numerous semi-supervised target recognition methods have emerged based on pseudo-labeling and consistency regularization, including Mixmatch [7], Fixmatch [8], and Mutexmatch [9]. The core idea of Mixmatch lies in mixing labeled data with unlabeled data, and subsequently training the model through diversity enhancement and consistency loss to enhance its generalization ability. Fixmatch sets a confidence threshold during pseudo-label learning, selecting unlabeled images with confidence scores exceeding this threshold for training. However, a significant proportion of unlabeled images with confidence scores falling below the threshold remain unused. To address this issue, Mutexmatch integrated complementary label (CoL) learning with Fixmatch, thus facilitating the exploitation of low-confidence unlabeled images. Drawing upon the aforementioned methods, recent advancements in semi-supervised SAR ATR [10], [11], [12] have emerged. For example, UA-Fixmatch [12] enhances the Fixmatch framework by incorporating the uncertainty of unlabeled data in selecting high-confidence images for pseudo-label learning. However, a substantial proportion of unlabeled images with low confidence remains underutilized.

To tackle the aforementioned challenges, we propose a novel semi-supervised SAR ATR method based on contrastive learning and CoL learning. First, drawing inspiration from Mutexmatch, we assign CoLs to low-confidence unlabeled images, utilizing the minimum prediction probability as a basis, thereby enhancing the utilization efficiency of these images. Notably, unlike Mutexmatch, we introduce an

Received 9 May 2024; revised 6 August 2024; accepted 4 September 2024. Date of current version 25 September 2024. This work was supported by the National Science Foundation of China under Grant U21B2039. *(Corresponding author: Lan Du.)*

The authors are with the National Key Laboratory of Radar Signal Processing, Xidian University, Xi'an 710071, China (e-mail: 1252377912@qq.com; dulan@mail.xidian.edu.cn; yuangd631@163.com).

Digital Object Identifier 10.1109/LGRS.2024.3458948

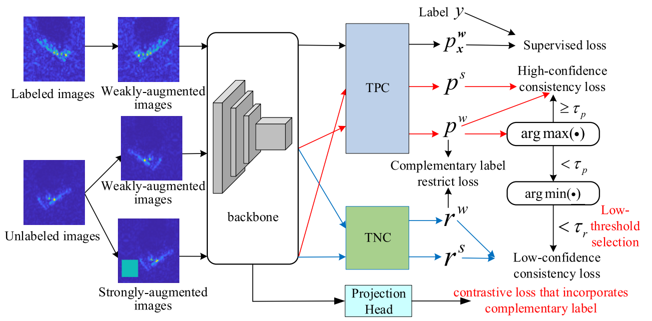

Fig. 1. Network structure of the proposed method, where $\boldsymbol{p}_n^w$ and $\boldsymbol{p}_n^s$ denote the prediction output of TPC for weakly augmented images and strongly augmented images, respectively. $\boldsymbol{r}_n^w$ and $\boldsymbol{r}_n^s$ denote the predicted output of TNC for weakly augmented images and strongly augmented images, respectively, and are two preset thresholds.

additional low-confidence threshold for filtering CoL, with the aim of mitigating the negative impact of erroneous CoL on model learning. Furthermore, we propose a contrastive loss that incorporates CoL. Compared to the traditional contrastive loss, our proposed contrastive loss leverages the unique properties of CoL to generate a more diverse set of negative sample pairs, thereby improving the recognition performance. Experiments on the MSTAR dataset demonstrate that our proposed method outperforms other semi-supervised recognition methods, especially under the condition of scarce labeled samples. The novelty of our proposed method can be summarized as follows.

1) We draw inspiration from CoL learning and further improve it within our network. By establishing a low-confidence threshold, our model can select more precise CoL, thereby mitigating the adverse impact of erroneous CoL on model learning.
2) We propose a contrastive loss that incorporates CoL, enabling the network to fully utilize low-confidence unlabeled images compared with the traditional contrastive loss.
3) Our proposed method achieves the SOTA semi-supervised recognition performance on the MSTAR dataset, achieving a recognition rate of 94.93% with only five labeled images per class.

II. Methodology

Fig. 1 depicts the architecture of the proposed method. The proposed model utilizes a backbone and two classifiers, i.e., the true-positive classifier (TPC) and the true-negative classifier (TNC), to predict both pseudo-labels and CoL. Both the TNC and TPC utilize two fully connected layers. Under semi-supervised conditions, the training data of a mini-batch comprises B labeled (l) images $\mathcal{X} = \{(\boldsymbol{x}_n^l, \boldsymbol{y}_n^l)\}_{n=1}^B$ and μB unlabeled (ul) images $\mathcal{U} = \{(\boldsymbol{x}_n^{ul})\}_{n=1}^{\mu B}$, where μ represents the relative proportion of $\mathcal{U}$ to $\mathcal{X}$ in the training data of a mini-batch. The labeled images are utilized for supervised learning, relying on the supervised loss. The unlabeled images are categorized into high-confidence and low-confidence images, depending on the predicted confidence scores derived from the TPC. Both high-confidence and low-confidence images contribute to the model training, subject to the constraints of high-confidence consistency loss (HCCL) and low-confidence consistency loss (LCCL), respectively. Meanwhile, we utilize the CoL learning loss to bring the category with the highest predicted confidence score in the TNC closer to the category with the lowest predicted score in the TPC. This loss serves to enhance the robustness of our model. Furthermore, to fully leverage the feature information of input images to assist semi-supervised learning, a projection head is appended to the backbone for contrastive learning. The projection head utilizes one fully connected layer, and the feature dimension within the embedding space is set to 256. The contrastive loss that incorporates CoL can be used to train the model simultaneously with both labeled and unlabeled images.

The subsequent section elaborates on the five losses: supervised loss, CoL learning loss, high confidence consistency loss, low confidence consistency loss, and contrastive loss that incorporates CoL.

A. Supervised Loss

The supervised loss is based on the true labels and utilizes the weakly augmented labeled images for model optimization. Specifically, the labeled images are transformed by weak augmentation first, i.e., the random horizontal flipping and random cropping, to obtain weakly augmented images. Subsequently, these weakly augmented images are fed into the backbone for feature extraction. Then, the TPC is utilized to produce the predicted probability vector. Finally, the cross-entropy loss function is employed for optimization, as formulated in the following equation:

$$L_{\text{sup}} = \frac{1}{B}\sum_{n=1}^{B} \text{CE}\left(\boldsymbol{y}_n^l, \mathcal{P}\left(\theta\left(\alpha_w\left(\boldsymbol{x}_n^l\right)\right)\right)\right) \tag{1}$$

where $\alpha_w(\boldsymbol{x}_n^l)$ represents the weakly augmented images, $\mathcal{P}(\theta(\alpha_w(\boldsymbol{x}_n^l)))$ represents the predicted probability vector. $\text{CE}(\cdot, \cdot)$ represents the cross-entropy loss function, which is expressed by the formula as follows:

$$\text{CE}(p, q) = -\sum p \ln q. \tag{2}$$

B. CoL Learning Loss

In our network, the TPC is utilized to predict the most probable class of the input image, whereas the TNC is utilized to predict the least probable class. To guarantee the robustness of the model, the class with the lowest score in the TPC's predicted probability vector should align with the class scoring the highest in the TNC's output probability vector. To fulfill this criterion, we incorporate a CoL learning loss, as formulated in the following equation:

$$L_{\text{cll}} = \frac{1}{\mu B}\sum_{n=1}^{\mu B} \text{CE}\left(\hat{\boldsymbol{q}}_n^w, \boldsymbol{r}_n^w\right) \tag{3}$$

where $\boldsymbol{r}_n^w$ represents the predicted probability vector of the weakly augmented image through TNC, and $\hat{\boldsymbol{q}}_n^w$ is the class corresponding to the minimum predicted probability of the weakly augmented image through TPC, i.e., the CoL. It is formulated as follows:

$$\hat{\boldsymbol{q}}_n^w = \text{argmin}\left(\boldsymbol{p}_n^w\right). \tag{4}$$

C. Consistency Loss

For unlabeled images, we first perform weak augmentation and strong augmentation, where strong augmentation includes adding gaussian noise and cutout [10]. Subsequently, the consistency loss is imposed to ensure that the prediction results of strongly augmented images align closely with those of weakly augmented images. Specifically, based on the predicted probability vector of TPC for weakly augmented images and two preset thresholds, τ_p and τ_r, the unlabeled images are categorized into high-confidence and low-confidence images. These two types of images are then trained separately, utilizing a HCCL and a LCCL, respectively. In the following, we will delve into the details of these two losses.

1) HCCL: First, the model selects the weakly augmented images with the highest probability scores exceeding τ_p, along with their corresponding strongly augmented images. Subsequently, this loss constrains the prediction outputs of TPC for the strongly augmented images to align with the pseudo-labels derived from the weakly augmented images. This process can be formulated as follows:

$$L_p = \frac{1}{\mu B}\sum_{n=1}^{\mu B}\mathbb{I}\left(\max\left(\boldsymbol{p}_n^w\right) \geq \tau_p\right)\mathrm{CE}\left(\hat{\boldsymbol{p}}_n^w, \boldsymbol{p}_n^s\right) \tag{5}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function, $\hat{\boldsymbol{p}}_n^w = \operatorname{argmax}(\boldsymbol{p}_n^w)$, $\hat{\boldsymbol{p}}_n^w$ and $\hat{\boldsymbol{p}}_n^s$ denote the prediction output of TPC for weakly augmented images and strongly augmented images, respectively.

2) LCCL: For images with the highest probability scores below the threshold τ_p, it is difficult for TPC to accurately predict their pseudo-labels. Nevertheless, predicting their CoL, i.e., the classes they are least likely to belong to, is easier. Therefore, we introduce a low-confidence threshold τ_r, enabling the model to assign CoL to images with low minimum probability scores. Notably, the setting of τ_r ensures a more accurate acquisition of CoL. Subsequently, in a similar manner to the constraint imposed by the HCCL, the low-confidence loss constrains the predicted output of TNC for strongly augmented images to align with the CoL derived from weakly augmented images. This process can be formulated as follows:

$$L_r = \frac{1}{\mu B}\sum_{n=1}^{\mu B}\mathbb{I}\left(\min\left(\boldsymbol{p}_n^w\right) < \tau_r\right)\mathrm{CE}\left(\boldsymbol{r}_n^w, \boldsymbol{r}_n^s\right) \tag{6}$$

where $\boldsymbol{r}_n^w$ and $\boldsymbol{r}_n^s$ denote the predicted output of TNC for weakly augmented images and strongly augmented images, respectively.

D. Contrastive Loss That Incorporates CoL

To fully exploit the feature information of input images to assist semi-supervised learning, we propose the contrastive loss that incorporates CoL. First, a projection head is appended to the backbone network, facilitating the mapping of input images into the embedding space. Subsequently, positive and negative sample pairs are constructed within this embedding space. Finally, the contrastive loss that incorporates CoL is computed, enabling the optimization of the network accordingly. Fig. 2 shows the schematic of the construction of positive and negative sample pairs. As can be seen from Fig. 2, for a sample with a true label or pseudo-label of class "A," samples sharing the same true label or pseudo-label of class "A" constitute positive sample pairs with it. Conversely, samples with true labels or pseudo-labels other than class "A" or having CoL of class "A" form negative sample pairs with it. After constructing positive and negative sample pairs, the cosine similarity function is employed to measure the similarity of sample pairs within the embedding space, which is expressed as follows:

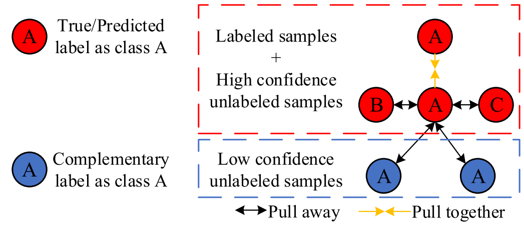

Fig. 2. Network structure of the proposed method.

$$\mathrm{cos\_sim}\left(x_i, x_j\right) = \frac{\left\langle e(x_i), e\left(x_j\right)\right\rangle}{\|e(x_i)\| \cdot \left\|e\left(x_j\right)\right\|} \tag{7}$$

where e denotes the embedding feature of the input image.

Based on the similarity between the sample pairs, the contrastive loss that incorporates CoL is constructed as follows:

$$L_c = \sum_{(x_i,x_j)\in P}\left(1 - \mathrm{cos\_sim}\left(x_i, x_j\right)\right)^2 + \sum_{(x_i,x_j)\in N}\left(1 + \mathrm{cos\_sim}\left(x_i, x_j\right)\right)^2 \tag{8}$$

where P denotes the positive sample pair set and N denotes the negative sample pair set.

Compared with the contrastive loss widely used in self-supervised learning, the proposed contrastive loss incorporates the CoL into the construction of negative sample pairs. This approach enriches the negative sample pairs and allows for a more comprehensive utilization of low-confidence samples, thereby contributing to a further improvement of recognition performance under limited labeled images. Notably, the proposed contrastive loss relies on a small amount of labeled data during training to construct positive and negative sample pairs. Therefore, it is more suitable for semi-supervised learning tasks rather than self-supervised learning tasks.

The total loss function of the entire network includes five weighted losses: supervised loss L_{sup}, CoL learning loss L_{cll}, HCCL L_p, LCCL L_r, and contrastive loss that incorporates CoL L_c

$$L_{\mathrm{total}} = L_{\mathrm{sup}} + \lambda_{\mathrm{cll}}L_{\mathrm{cll}} + \lambda_p L_p + \lambda_r L_r + \lambda_c L_c \tag{9}$$

where λ_{cll}, λ_p, λ_r, and λ_c denote the weights of corresponding losses.

III. Experimental Results and Analysis

A. Dataset

The dataset used in our experiments is the MSTAR measured dataset [14]. The MSTAR dataset, acquired by radar

TABLE I

TRAINING AND TESTING IMAGES FOR THE TEN-TARGET DATA

No.	Type	Training (17°)	Testing (15°)
1	BMP2	233	196
2	BTR70	233	196
3	T72	232	196
4	BTR60	255	195
5	2S1	299	274
6	BRDM2	298	274
7	D7	299	274
8	T62	299	273
9	ZIL131	299	274
10	ZIL234	299	274

operating in X-band, contains ten different classes of ground military targets with a resolution of 0.3 $\times$ 0.3 m. We employ the ten-class MSTAR data under standard operating conditions. This is the widely recognized configurations for semi-supervised SAR ATR [11], [12], [13]. For the original complex SAR data in MSTAR, we sequentially apply modulus operation, central cropping to extract sized SAR image chips, and then normalize the resulting images, which subsequently serve as inputs to our network. The training and testing images on the ten-target data is shown in Table I.

B. Experiment Setup

In the semi-supervised recognition experiments, we employed the WideResNet [15], specifically with a depth of 28 and a width of 2, as the backbone. This backbone structure is commonly utilized in semi-supervised experiments. WideResNet can guarantee high model performance without significantly increasing the computational cost by increasing the width of the network. In the loss function, the weight of different loss items λ_{cll}, λ_p, λ_r, and λ_c are set to 1, 1, 1, and 0.1, respectively. τ_p and τ_r are set to 0.95 and 0.05, respectively. Batchsize is set to 32 and μ is set to 7. Both the TNC and TPC utilize two fully connected layers. The projection head utilizes one fully connected layer, and the feature dimension within the embedding space is set to 256. The stochastic gradient descent algorithm is adopted for optimization. The total number of training iterations is fixed at 100 000. Three different partitions of labeled and unlabeled images were utilized for our experiments. The recognition results were obtained by calculating the mean and standard deviation of recognition accuracies across these three partitions.

C. Semi-Supervised Recognition Experiments

In the experiment, nine mainstream semi-supervised target recognition methods are employed for comparison with the proposed method, including MGAN-CNN [16], SCA [11], MBM [12], Fixmatch [8], Maxmatch [17], Mutexmatch [9], CSSL [18], NP-match [19], and UA-Fixmatch [13]. Recognition accuracies of various methods are presented in Table II, with each method evaluated under conditions of 5, 10, 20, and 40 labeled images per class.

TABLE II

RECOGNITION ACCURACIES OF VARIOUS METHODS

Number of labeled images per class	5	10	20	40
MGAN-CNN[15]	-	-	85.23	90.82
SCA[11]	-	95.22±3.18	98.23±0.09	99.85±0.10
MBM[12]	-	97.21±2.44	99.56±0.16	99.58±0.07
Fixmatch[8]	70.10±5.23	94.72±1.36	97.24±0.65	99.26±0.52
Maxmatch[17]	68.99±5.09	96.04±0.96	98.62±1.23	99.57±0.38
Mutexmatch[9]	87.23±2.52	97.18±0.87	99.18±0.33	99.80±0.01
CSSL[18]	73.85±5.02	95.92±1.13	99.42±0.28	99.59±0.08
NP-Match[19]	69.18±3.82	96.72±0.97	98.65±0.24	99.26±0.02
UA-Fixmatch[13]	88.65±3.26	98.58±1.02	99.84±0.11	99.86±0.08
Proposed method	**94.93±2.50**	**99.41±0.47**	**99.88±0.07**	**99.92±0.04**

TABLE III

ABLATION EXPERIMENTS ON THE CONTRASTIVE LOSS AND LTS

Contrastive loss		LTS	Number of labeled images per class	
w/o CoL	w/ CoL		5	10
×	×	×	87.23	97.18
√	×	×	89.04	98.72
×	√	×	93.69	99.09
×	×	√	89.08	98.93
√	×	√	91.63	99.18
×	√	√	**94.93**	**99.63**

As can be seen from Table II, the proposed method attains the highest recognition accuracy across four distinct experimental settings. Notably, when each class contains only five labeled images, the performance advantage of the proposed method is more prominent. When each class contains ten labeled images, the proposed method achieves 99.63% recognition accuracy, representing a 1.05% improvement over the best result of the comparison method. Compared with SCA and MBM, UA-Fixmatch exhibits superior recognition performance across various numbers of labeled images. However, UA-Fixmatch still has a gap in recognition performance compared with the proposed method. Therefore, the proposed method achieves SOTA semi-supervised SAR target recognition performance on the MSTAR dataset.

D. Ablation Experiment

Compared to Mutexmatch, the proposed method incorporates two additional modules: low-threshold selection (LTS) for low-confidence images and contrastive loss that incorporates CoL. To analyze the effects of these two modules, we conducted ablation experiments under the conditions of five and ten labeled images per class. The experimental results are presented in Table III.

When the contrastive loss and LTS are excluded, the proposed method reverts to Mutexmatch. Combining Mutexmatch with the contrastive loss, the recognition accuracies have been significantly improved. Notably, by incorporating CoL into the contrastive loss to construct a more diverse set of negative sample pairs, the improvement in recognition accuracy is even more pronounced, as demonstrated in rows 4 and 5 as well as rows 7 and 8. It demonstrates the effectiveness of incorporating CoL into the contrastive loss. Compared with Mutexmatch, our

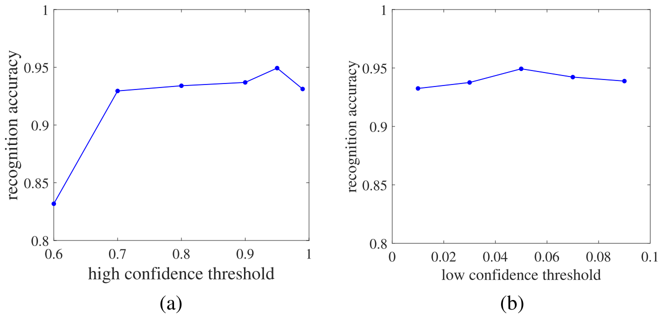

Fig. 3. Influence of the high-confidence threshold and low-confidence threshold on recognition performance. (a) High-confidence threshold. (b) Low-confidence threshold.

proposed method adopts LTS to utilize low-confidence images more accurately, and the recognition accuracies increased by 1.85% and 1.75% under the conditions of five and ten labeled images per class, respectively. When LTS and contrastive loss that incorporates CoL are simultaneously integrated, the proposed method attains the best recognition performance. It proves that these two modules can improve recognition performance under conditions of limited labeled images.

E. Influence of the High-Confidence Threshold and Low-Confidence Threshold

To study the influence of the values of high-confidence threshold τ_p and low-confidence threshold τ_r on recognition performance, we set multiple different values for τ_p and τ_r, respectively, and statistically analyzed the corresponding recognition accuracy. When each class contains five labeled images, Fig. 3 shows the experimental results.

As shown in Fig. 3(a), the optimal recognition performance is achieved when the value of τ_p is set to 0.95. When the value of τ_p is relatively small, it is more likely to encounter pseudo-label errors in high-confidence images, which negatively impacts the model training. Conversely, a higher value of τ_p results in a smaller number of high-confidence images, which makes the model make insufficient use of high-confidence images. From Fig. 3(b), we can observe that the optimal recognition performance is achieved when the value of τ_r is set to 0.05. When the value of τ_r is relatively small, it means that only a few low-confidence images are selected, resulting in insufficient utilization of these samples by the network and subsequently poor performance. Conversely, a higher value of τ_r increases the likelihood of incorrect CoL for low-confidence images, leading to a decline in recognition performance.

IV. Conclusion

In this letter, a semi-supervised SAR target recognition network based on contrastive learning and CoL learning is proposed, which significantly improves the target recognition performance under limited labeled images. On the one hand, the model uses CoL learning to make full use of low-confidence images. And by establishing a low-confidence threshold, we mitigate the adverse effects caused by incorrect CoL, thereby enabling a more accurate utilization of low-confidence images. On the other hand, we propose the contrastive loss that incorporates CoL. Compared to traditional contrastive learning losses, this loss enriches the negative sample pairs, enabling a more comprehensive utilization of low-confidence images. Experiments on the MSTAR dataset demonstrate that the proposed method attains the SOTA semi-supervised SAR target recognition performance. In our future work plans, we also consider applying the ideas of contrastive learning and CoL learning to semi-supervised SAR target detection tasks.

References

[1] S. Zhao, Z. Zhang, T. Zhang, W. Guo, and Y. Luo, "Transferable SAR image classification crossing different satellites under open set condition," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[2] Y. Chen, M. Mancini, X. Zhu, and Z. Akata, "Semi-supervised and unsupervised deep visual learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 3, pp. 1327–1347, Mar. 2024.

[3] X. Yang, Z. Song, I. King, and Z. Xu, "A survey on deep semi-supervised learning," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 9, pp. 8934–8954, Sep. 2023.

[4] D.-H. Lee, "Pseudo-Label: The simple and efficient semi-supervised learning method for deep neural networks," in *Proc. Workshop Challenges Represent. Learn.*, 2013, vol. 3, no. 2, p. 896.

[5] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. Int. Conf. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1195–1204.

[6] Q. Xie, Z. Dai, E. Hovy, M.-T. Luong, and Q. V. Le, "Unsupervised data augmentation for consistency training," in *Proc. NIPS*, vol. 33, 2020, pp. 6256–6268.

[7] D. Berthelot, N. Carlini, I. Goodfellow, N. Papernot, A. Oliver, and C. A. Raffel, "MixMatch: A holistic approach to semi-supervised learning," in *Proc. 33rd Int. Conf. Neural Inf. Process. Syst.*, 2019, pp. 5049–5059.

[8] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2020, pp. 596–608.

[9] Y. Duan et al., "MutexMatch: Semi-supervised learning with mutex-based consistency regularization," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 6, pp. 8441–8455, Jun. 2024.

[10] T. DeVries and G. W. Taylor, "Improved regularization of convolutional neural networks with cutout," 2017, *arXiv:1708.04552*.

[11] C. Wang et al., "Semisupervised learning-based SAR ATR via self-consistent augmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 4862–4873, Jun. 2021.

[12] Y. Tian, J. Sun, P. Qi, G. Yin, and L. Zhang, "Multi-block mixed sample semi-supervised learning for SAR target recognition," *Remote Sens.*, vol. 13, no. 3, p. 361, Jan. 2021.

[13] X. Zhang, Y. Luo, and L. Hu, "Semi-supervised SAR ATR via epoch- and uncertainty-aware pseudo-label exploitation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5209015.

[14] T. D. Ross, S. W. Worrell, and V. J. Velten, J. C. Mossing, and M. L. Bryant, "Standard SAR ATR evaluation experiments using the MSTAR public release dataset," *Proc. SPIE*, vol. 3370, pp. 566–573, Sep. 1998.

[15] S. Zagoruyko and N. Komodakis, "Wide residual networks," 2016, *arXiv:1605.07146*.

[16] X. Liu et al., "Semi-supervised SAR ATR via conditional generative adversarial network with multi-discriminator," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Brussels, Belgium, Jul. 2021, pp. 2361–2364.

[17] Y. Jiang et al., "MaxMatch: Semi-supervised learning with worst-case consistency," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5970–5987, May 2023.

[18] J. Lienen and E. Hüllermeier, "Credal self-supervised learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 14370–14382.

[19] J. Wang, T. Lukasiewicz, D. Massiceti, X. Hu, V. Pavlovic, and A. Neophytou, "NP-Match: When neural processes meet semi-supervised learning," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 22919–22934.